Srikrishnaa Vadivel

Computational Physicist | First Principles | HPC | Scientific Software

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Summary

Computational physics PhD researcher focused on first-principles calculations of self-trapped excitons, scientific software, and high-performance simulation. Experience includes leadership-class HPC, PETSc/SLEPc, MPI/OpenMP, Python/C++, and ML-assisted materials workflows.

Research

Yale University, PhD Researcher 2021–present

- Perform first-principles calculations of self-trapped excitons using HPC resources at Oak Ridge, Aurora, LBNL, and LLNL.
- Build reproducible Linux, Bash, Python, C/C++, CMake, PETSc/SLEPc, MPI/OpenMP, Docker, and Kubernetes workflows for large simulations.
- Explore machine-learning surrogates and equivariant molecular models for materials and molecular systems.

Selected Technical Projects

- **DOLFINx PDE Gallery**: weak forms for Poisson, heat, and elasticity, plus solver/visualization scaffold.
- **First-Principles LiF**: plane-wave pseudopotential Hamiltonian and band plotting.
- **Physics Action Notes**: path integrals, imaginary time, EM, GR, QED/QCD, Standard Model, and strings from action principles.
- **Meep Ring Resonator**: photonics FDTD geometry and source setup.

Industry

Probe Technologies / Weatherford 2019–2021

Built CUDA/OpenGL waveform processing and visualization software for acoustic logging tools; also delivered embedded sensor logging firmware and FPGA logic.

Education

Yale University, PhD Electrical Engineering, 2021–present
Cornell University, MEng ECE, 2017–2018
Cornell University, BA ECE, 2013–2017

Skills

DFT workflows, PDEs, numerical linear algebra, PETSc, SLEPc, MPI, OpenMP, Python, C++, CMake, Linux, CUDA, PyTorch, DOLFINx concepts, Meep concepts.